

Category	Item	Value	Unit
Material	Concrete	100	m³
	Steel	50	kg
	Brick	200	kg
	Wood	30	m³
Labor	Construction Worker	10	hr
	Electrician	5	hr
	Plumber	3	hr
	Painter	2	hr
Equipment	Excavator	1	hr
	Truck	2	hr
	Generator	1	hr
	Drill	1	hr

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9702/36

October/November 2012

2 hours

Additional Materials: As listed in the Confidential Instructions.

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO NOT WRITE IN ANY BARCODES.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] after each question or part question.

For Examiner's Use	
1	
2	
Total	



[Turn over

You may not need to use all of the materials provided.

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- 1** In this experiment, you will investigate how the light intensity incident on a light sensor varies with distance from the light source.

- (a)** You are provided with a lamp mounted inside one end of a black paper tube, together with a light sensor in the form of a light-dependent resistor (LDR) mounted on a half-metre rule, as shown in Fig. 1.1.

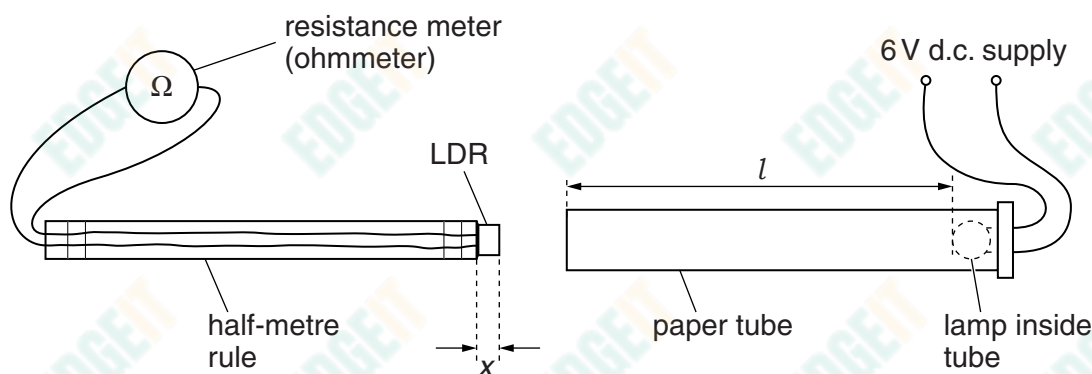


Fig. 1.1

- (b) (i)** Measure and record the distance l from the lamp to the open end of the tube, as shown in Fig. 1.1.

$l =$

- (ii)** Measure and record the distance x from the front of the LDR to the end of the half-metre rule, as shown in Fig. 1.1.

$x =$ [1]

- (c) (i) Push the rule into the tube until the LDR is approximately half way along the tube, as shown in Fig. 1.2.

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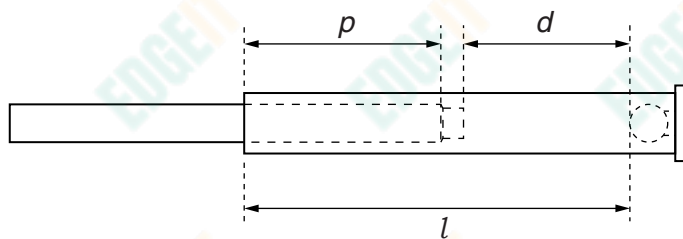


Fig. 1.2

- (ii) Record the length p of the half-metre rule inside the tube.

$p =$

- (iii) Calculate the distance d of the LDR from the lamp using $d = l - (x + p)$.

$d =$

- (iv) Switch on the lamp and record the resistance reading R on the resistance meter.

$R =$ $\text{k}\Omega$ [1]

☐

- (d) Repeat (c) using different values of p until you have six sets of values of p and R for $p \leq 20$ cm.

In your table of results include values for d and $d^{1.5}$ ($d^{1.5} = \sqrt{d^3}$).

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[10]

- (e) (i) Plot a graph of R on the y -axis against $d^{1.5}$ on the x -axis.

[3]

- (ii) Draw the straight line of best fit.

[1]

- (iii) Determine the gradient and y -intercept of this line.

gradient =

y -intercept =

[2]

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- (f) The quantities R and d are related by the equation

$$R = ad^{1.5} + b$$

where a and b are constants.

Using your answers from (e)(iii), determine the values of a and b .
Give appropriate units.

$a =$

$b =$

[2]

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2 In this experiment, you will investigate the speed of a cylindrical piston moving through water.

- (a) (i)** You are provided with a container of water and a piece of modelling clay which can be moulded into a cylindrical piston. You are also provided with a card on which is written the internal diameter d_0 of the container, together with two other values A and B .

Mould the piece of modelling clay into a cylinder with a diameter approximately equal to value A .

Measure and record the diameter d of this cylinder.

$d = \dots\dots\dots$ [1]

- (ii)** Estimate the percentage uncertainty in your value of d .

percentage uncertainty = $\dots\dots\dots$ [1]

- (iii)** You are also provided with a string running over a pulley. The string has a mass X attached to one end and a paperclip at the other end.

Push the paperclip into the centre of your clay cylinder and lower it into the container of water, as shown in Fig. 2.1, with the clay cylinder axis vertical.

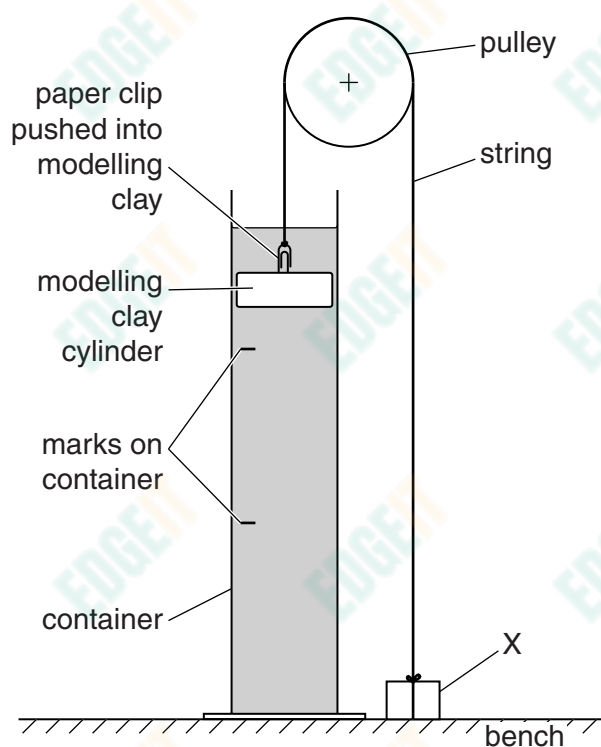


Fig. 2.1

- (iv) Measure and record the distance h between the marks on the container of water.

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$$h = \dots\dots\dots [1]$$

- (b) (i) Raise X until the clay cylinder reaches the bottom of the container of water. Release X and take measurements to determine the time t taken for the clay cylinder to rise from the lower mark to the higher mark.

$$t = \dots\dots\dots [2]$$

- (ii) Calculate the average speed v of the clay cylinder between the marks using $v = \frac{h}{t}$.

$$v = \dots\dots\dots [1]$$

- (iii) Justify the number of significant figures that you have given for your value of v .

.....
.....
..... [1]

- (c) (i) Remove the modelling clay from its paperclip and re-mould it into a cylinder with a diameter approximately equal to value B from the card. Measure and record the diameter d of the cylinder.

$$d = \dots\dots\dots [1]$$

- (ii) Push the paperclip into the centre of your clay cylinder again and lower it into the container of water. Repeat **(b)(i)** and **(b)(ii)**.

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$t =$

$v =$

[2]

- (d) (i) It is suggested that the relationship between v and d is

$$v = k (d_0 - d)$$

where k is a constant and d_0 is given on your card.

Using your data, calculate two values of k .

first value of $k =$

second value of $k =$

[1]

- (ii) Explain whether your results support the suggested relationship.

.....

 [1]

- (e) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

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1.

2.

3.

4.

[4]

- (ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.

2.

3.

4.

[4]

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